

Partial Exam	CT1000S Structural Mechanics, Stability
Total number of pages	8 pages (excl cover)
Date and time	31 JAN 2025 from 09:00-12:00
Responsible lecturer	J.W. Welleman

Only the work / answers written on examination paper will be assessed, unless otherwise specified under 'Additional Information'.

Exam questions (to be filled in by course examiner)

Total number of questions: 2

☒ **questions may differ in weight** (the time mentioned is an indicator for the weight)

Use of tools and sources of information (to be filled in by course examiner)

Not allowed:

- Mobile phone, smart Phone or devices with similar functions.
- Answers written with red pen or with pencils.
- Calculators with CAS and/or wifi/BT and/or PDF capabilities
- Tools and/or sources of information unless otherwise specified below.

Allowed:

- ☐ **books** ☐ **notes** ☐ **dictionaries** ☐ **syllabus**
- ☐ **formula sheets (see also below under 'additional information')** ☒ **calculators**
- ☐ **computer** ☐ **...**
- ☒ **scientific (graphical)calculator** ☒ **drawing material**

Additional information (if necessary to be filled in by the examiner)

- **Use for each problem a separate examination paper**
- **The question form contains fomula sheets which can be used.**
- **Students can take the question form home after the exam.**
- **Specify the correct BSc or MSc course code on the exam paper**
- **BSc students are allowed to answer in Dutch**
- **No student leaves without delivering an exam paper with a name on it!**

Exam graded by: (the marking period is 15 working days at most)



Every suspicion of fraud is reported to
the Board of Examiners.

Mobile Phone
OFF.

Problem 1 : Theory**(approx. 20 min)**

A stability assessment is required for the structure presented in fig. 1. The connections in both C and D are hinged and r is a linear elastic rotational spring. Only column AC can bend, all other members are rigid. Any axial deformation is assumed to be negligible.

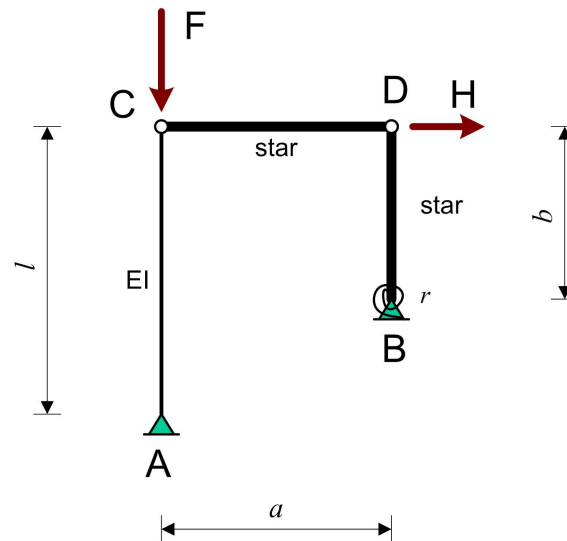


Figure 1 : Structure with concentrated forces

NOTE:

Modelling involves a sketch of the model, a description (formula) of possible springs used and or boundary conditions needed.

Questions:

- Sketch the expected mode(s) for the limit of a stable equilibrium.
- Simplify this structural model to a mathematical model for which the limit load can be found.
- Explain the steps involved to find the limit load. You do not need to solve for the expression of the answer.

Problem 2 : Buckling and 2nd order effect**(approx. 40 min)**

A frame structure is loaded with two concentrated vertical forces F as is shown in fig. 2. Apart from this vertical load also a horizontal load H is applied. Bending stiffnesses expressed in EI can be depicted from fig. 2 and worth to mention that the upper and lower beams in the frame have different bending stiffnesses. The length l of the structure is 10 m, the height h is 5 m. The influence of any axial deformation can be neglected in the problem and all connections between columns and beams are assumed to be rigid.

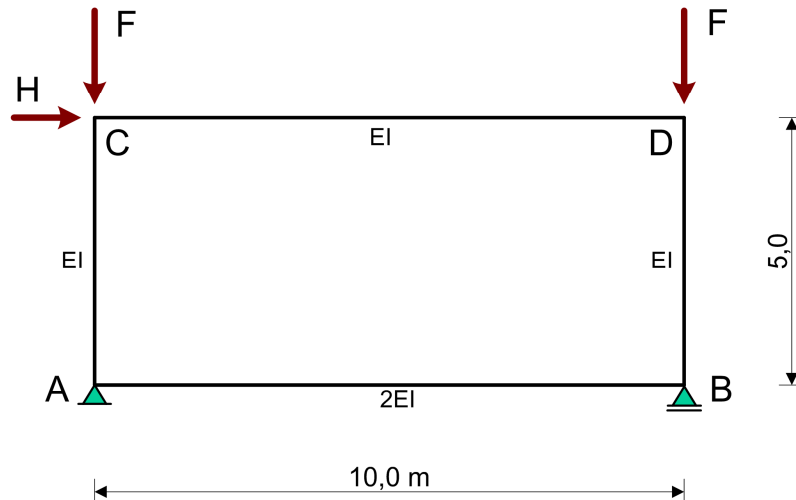


Figure 2 : Frame loaded with concentrated forces

Given: $EI = 15000 \text{ kNm}^2$; $F = 1000 \text{ kN}$; $H = 90 \text{ kN}$;

From a *geometrical linear analysis* is known that the maximum moment in column AC is given as:

$$M_{\max}^{AC} = \frac{(l + 3h)}{3(l + 4h)} Hh$$

Also from this analysis is known that the first order horizontal displacement at C can be expressed as:

$$u_{C-1} = \frac{(l^2 + 6hl + 6h^2)}{36(l + 4h)} \frac{Hh^2}{EI}; \quad (\text{check this at home if you like ...})$$

Questions:

- Sketch the buckling mode and explain how to find the buckling load for this problem.
- Find the buckling load in [kN] for the given situation and support your approach if needed with sketches.
- Find the first order moment in A and C in [kNm].
- Find the second order displacement in [m] of point C for the given F and H
- Find the second order moment in A in [kNm] for the given F and H .

FORMULAS

Eulerian Buckling:

$$F_k = \frac{\pi^2 EI}{l_k^2}$$

$$\frac{1}{F_k} = \frac{1}{r} + \frac{1}{\frac{\pi^2 EI}{4l^2}} \Rightarrow l_k = l \sqrt{4 + \frac{10}{\rho}}$$

met: $\rho = \frac{rl}{EI}$

$$F_k = \frac{(\eta_1 + \eta_2)^2}{\eta_1 \eta_2 (\eta_1 + \eta_2 - 4)} \times \frac{\pi^2 EI}{l^2} \quad \text{with:} \quad \begin{aligned} \eta_1 &= 4 + \frac{10}{\rho_1}; \rho_1 = \frac{r_1 l}{EI} \\ \eta_2 &= 4 + \frac{10}{\rho_2}; \rho_2 = \frac{r_2 l}{EI} \end{aligned}$$

$$F_k = \frac{(5 + 2\rho_1)(5 + 2\rho_2)}{(5 + \rho_1)(5 + \rho_2)} \cdot \frac{\pi^2 EI}{l^2} \quad \text{with:} \quad \rho_1 = \frac{r_1 l}{EI} \quad \rho_2 = \frac{r_2 l}{EI}$$

Gonio results:

$$\tan \alpha l = \alpha l \Rightarrow \alpha l \cong 4.49 \cong \frac{\pi}{0.7}$$

Deformation energy:

$$E_v = \int \frac{1}{2} EA \varepsilon^2 dx \quad (\text{extension})$$

$$E_v = \int \frac{1}{2} EI \kappa^2 dx \quad (\text{bending})$$

Complementary energy:

$$E_c = \int \frac{N^2}{2EA} dx \quad (\text{extension})$$

$$E_c = \int \frac{M^2}{2EI} dx \quad (\text{bending})$$

Rayleigh:

$$F_k = \frac{E_v}{\int \frac{1}{2} \left(\frac{dw}{dx} \right)^2 dx}$$

Kinematic and Constitutive relations:

$$\varphi = -\frac{dw}{dx} \quad \kappa = \frac{d\varphi}{dx} \quad M = EI \kappa$$

Differential equation:

$$w'''' + \alpha^2 w'' = 0 \quad \text{with:} \quad \alpha^2 = \frac{F}{EI}$$

$$w(x) = C_1 + C_2 x + C_3 \cos \alpha x + C_4 \sin \alpha x$$

$$S_z(x) = M' - Fw' \quad \text{with:} \quad S_z(x) = -F \cdot C_2$$

Kinematic relations:

$$\varepsilon = \frac{du}{dx}$$

$$\kappa = -\frac{d^2 w}{dx^2}$$

Constitutive relations:

$$N = EA \cdot \varepsilon$$

$$M = EI \cdot \kappa$$

Math tools:

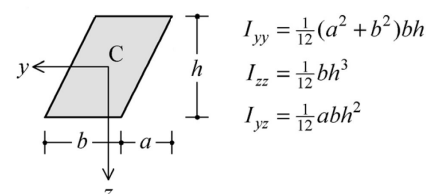
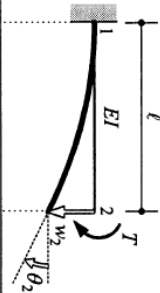
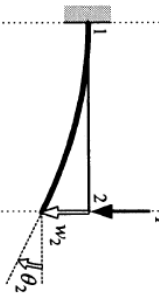
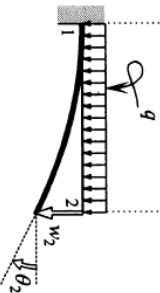
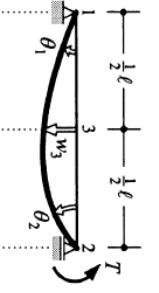
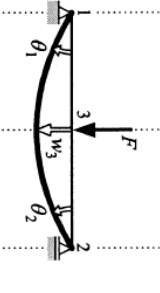
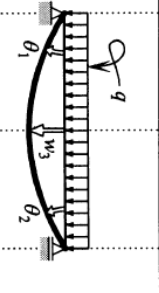
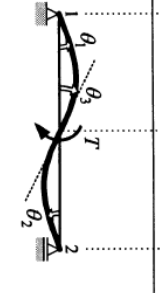


Figure	Area, coordinates centroid C	Second moments of area	
		centroidal	other
	Rectangle $A = bh$ $\bar{y}_C = \frac{1}{2}b$ $\bar{z}_C = \frac{1}{2}h$	$I_{yy} = \frac{1}{12}b^3h$ $I_{zz} = \frac{1}{12}bh^3$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = \frac{1}{3}b^3h$ $I_{\bar{z}\bar{z}} = \frac{1}{3}bh^3$ $I_{\bar{y}\bar{z}} = \frac{1}{4}b^2h^2$
	Parallelogram $A = bh$ $\bar{y}_C = \frac{1}{2}(a+b)$ $\bar{z}_C = \frac{1}{2}h$	$I_{yy} = \frac{1}{12}(a^2+b^2)bh$ $I_{zz} = \frac{1}{12}bh^3$ $I_{yz} = \frac{1}{12}ab h^2$	$I_{\bar{z}\bar{z}} = \frac{1}{3}bh^3$
	Triangle $A = \frac{1}{2}bh$ $\bar{y}_C = \frac{1}{3}(2a-b)$ $\bar{z}_C = \frac{2}{3}h$	$I_{yy} = \frac{1}{36}(a^2-ab+b^2)bh$ $I_{zz} = \frac{1}{36}bh^3$ $I_{yz} = \frac{1}{72}(2a-b)bh^2$	$I_{\bar{z}\bar{z}} = \frac{1}{4}bh^3$ $I_{\bar{y}\bar{z}} = \frac{1}{8}(2a-b)bh^2$ $I_{\bar{z}\bar{z}} = \frac{1}{12}bh^3$
	Trapezium $A = \frac{1}{2}(a+b)h$ $\bar{z}_C = \frac{1}{3}\frac{a+2b}{a+b}h$	$I_{zz} = \frac{1}{36}\frac{a^2+4ab+b^2}{a+b}h^3$	$I_{\bar{z}\bar{z}} = \frac{1}{12}(a+3b)h^3$ $I_{\bar{z}\bar{z}} = \frac{1}{12}(3a+b)h^3$
	Circle $A = \pi R^2$	$I_{yy} = I_{zz} = \frac{1}{4}\pi R^4$ $I_{yz} = 0$ $I_p = \frac{1}{2}\pi R^4$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{5}{4}\pi R^4$ $I_{\bar{y}\bar{z}} = \pi R^4$

Figure	Area, coordinates centroid C	Second moments of area	
		centroidal	other
	Thick-walled ring $A = \pi(R_2^2 - R_1^2)$	$I_{yy} = I_{zz} = \frac{1}{4}\pi(R_2^4 - R_1^4)$ $I_{yz} = 0$ $I_p = \frac{1}{2}\pi(R_2^4 - R_1^4)$	
	Thin-walled ring $A = 2\pi R t$	$I_{yy} = I_{zz} = \pi R^3 t$ $I_{yz} = 0$ $I_p = 2\pi R^3 t$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = 3\pi R^3 t$
	Semicircle $A = \frac{1}{2}\pi R^2$ $\bar{y}_C = 0$ $\bar{z}_C = \frac{4}{3\pi}R$ $= 0.424R$	$I_{yy} = \frac{1}{8}\pi R^4 = 0.393R^4$ $I_{zz} = (\frac{8}{\pi} - \frac{8}{9\pi})R^4$ $= 0.110R^4$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{1}{8}\pi R^4$ $I_{\bar{y}\bar{z}} = 0$
	Semicircular ring $A = \pi R t$ $\bar{y}_C = 0$ $\bar{z}_C = \frac{2}{\pi}R$ $= 0.637R$	$I_{yy} = \frac{1}{2}\pi R^3 t$ $I_{zz} = (\frac{\pi}{2} - \frac{4}{\pi})R^3 t$ $= 0.298R^3 t$ $I_{yz} = 0$	$I_{\bar{y}\bar{y}} = I_{\bar{z}\bar{z}} = \frac{1}{2}\pi R^3 t$ $I_{\bar{y}\bar{z}} = 0$

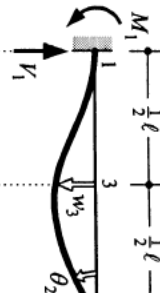
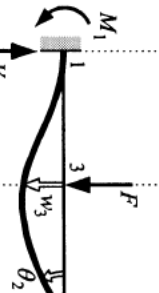
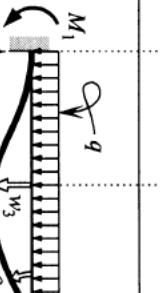
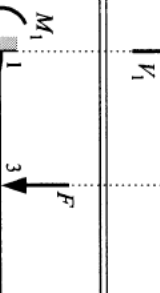
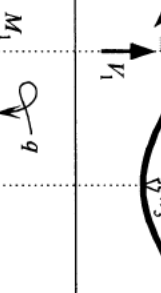
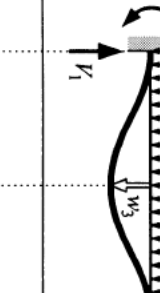
(1)		$\theta_2 = \frac{T\ell}{EI}; w_2 = \frac{T\ell^2}{2EI}$
(2)		$\theta_2 = \frac{F\ell^2}{2EI}; w_2 = \frac{F\ell^3}{3EI}$
(3)		$\theta_2 = \frac{q\ell^3}{6EI}; w_2 = \frac{q\ell^4}{8EI}$
(4)		$\theta_1 = \frac{1}{6} \frac{T\ell}{EI}; \theta_2 = \frac{1}{3} \frac{T\ell}{EI}; w_3 = \frac{1}{16} \frac{T\ell^2}{EI}$
(5)		$\theta_1 = \theta_2 = \frac{1}{16} \frac{F\ell^2}{EI}; w_3 = \frac{1}{48} \frac{F\ell^3}{EI}$
(6)		$\theta_1 = \theta_2 = \frac{1}{24} \frac{q\ell^3}{EI}; w_3 = \frac{5}{384} \frac{q\ell^4}{EI}$
(a)		$\theta_1 = \theta_2 = \frac{1}{24} \frac{T\ell}{EI}; \theta_3 = \frac{1}{12} \frac{T\ell}{EI}; w_3 = 0$

simply supported beam (statically determinate)

forget-me-nots

statically indeterminate beam (two fixed ends)

statically indeterminate beam (one fixed end)

(7)		$\theta_2 = \frac{1}{4} \frac{T\ell}{EI}; w_3 = \frac{1}{32} \frac{T\ell^2}{EI}$ $M_1 = \frac{1}{2} T; V_1 = V_2 = \frac{3}{2} T$
(8)		$\theta_2 = \frac{1}{32} \frac{F\ell^2}{EI}; w_3 = \frac{7}{768} \frac{F\ell^3}{EI}$ $M_1 = \frac{3}{16} F\ell; V_1 = \frac{11}{16} F; V_2 = \frac{5}{16} F$
(9)		$\theta_2 = \frac{1}{48} \frac{q\ell^3}{EI}; w_3 = \frac{1}{192} \frac{q\ell^4}{EI}$ $M_1 = \frac{1}{8} q\ell^2; V_1 = \frac{5}{8} q\ell; V_2 = \frac{3}{8} q\ell$
(10)		$w_3 = \frac{1}{192} \frac{F\ell^3}{EI}$ $M_1 = M_2 = \frac{1}{8} F\ell; V_1 = V_2 = \frac{1}{2} F$
(11)		$w_3 = \frac{1}{384} \frac{q\ell^4}{EI}$ $M_1 = M_2 = \frac{1}{12} q\ell^2; V_1 = V_2 = \frac{1}{2} q\ell$
(b)		$\theta_3 = \frac{1}{16} \frac{T\ell}{EI}; w_3 = 0$ $M_1 = M_2 = \frac{1}{4} T; V_1 = V_2 = \frac{3}{2} T$

	$\theta_1 = \frac{Fb\ell(\ell+b)}{6EI} = \frac{F\ell^2}{6EI} \left(2\frac{a}{\ell} - 3\frac{a^2}{\ell^2} + \frac{a^3}{\ell^3} \right)$ $\theta_2 = \frac{Fb\ell(\ell+a)}{6EI} = \frac{F\ell^2}{6EI} \left(\frac{a}{\ell} - \frac{a^3}{\ell^3} \right)$
	$M_1 = \frac{Fb\ell(\ell^2-b^2)}{2\ell^2} = F\ell \left(\frac{a}{\ell} - \frac{3a^2}{2\ell^2} + \frac{1a^3}{2\ell^3} \right)$ $V_1 = \frac{Fb(3\ell^2-b^2)}{2\ell^3} = F \left(1 - \frac{3a^2}{2\ell^2} + \frac{1a^3}{2\ell^3} \right)$ $V_2 = \frac{Fa^2(3\ell-a)}{2\ell^3} = F \left(\frac{3a^2}{2\ell^2} - \frac{1a^3}{2\ell^3} \right)$ $\theta_2 = \frac{Fa^2b}{4EI\ell} = \frac{F\ell^2}{4EI} \left(\frac{a^2}{\ell^2} - \frac{a^3}{\ell^3} \right)$
	$M_1 = \frac{Fb\ell^2}{\ell^2} = F\ell \left(\frac{a}{\ell} - 2\frac{a^2}{\ell^2} + \frac{a^3}{\ell^3} \right)$ $V_1 = \frac{Fb\ell(\ell+2a)}{\ell^3} = F \left(1 - 3\frac{a^2}{\ell^2} + 2\frac{a^3}{\ell^3} \right)$ $M_2 = \frac{Fa^2b}{\ell^2} = F\ell \left(\frac{a^2}{\ell^2} - \frac{a^3}{\ell^3} \right)$ $V_2 = \frac{Fa^2(\ell+2b)}{\ell^3} = F\ell \left(\frac{a^2}{\ell^2} - 2\frac{a^3}{\ell^3} \right)$
	$M_1 = \frac{3EI}{\ell^2} w^0; \quad V_1 = V_2 = \frac{3EI}{\ell^3} w^0$ $\theta_2 = \frac{3}{2} \frac{w^0}{\ell}$ $\theta_3 = \frac{9}{8} \frac{w^0}{\ell}; \quad w_3 = \frac{5}{16} w^0$
	$M_1 = M_2 = \frac{6EI}{\ell^2} w^0; \quad V_1 = V_2 = \frac{12EI}{\ell^3} w^0$ $\theta_3 = \frac{3}{2} \frac{w^0}{\ell}; \quad w_3 = \frac{1}{2} w^0$

settlements

support reactions and rotations at the beam ends

properties of plane figures to be used for the moment-area theorems

	rectangle: $y = h$ $A = bh$ $x_C = \frac{1}{2}b$
	triangle: $y = h \left\{ 1 - \frac{x}{b} \right\}$ $A = \frac{1}{2}bh$ $x_C = \frac{1}{3}b$
	parabola: $y = h \left\{ 1 - \left(\frac{x}{b} \right)^2 \right\}$ $A = \frac{1}{3}bh$ $x_C = \frac{1}{4}b$
	parabola: $y = h \left\{ 1 - \left(\frac{x}{b} \right)^2 \right\}$ $A = \frac{2}{3}bh$ $x_C = \frac{3}{8}b$
	parabola: $A = \frac{2}{3}bh$ $x_C = \frac{1}{2}b$
	trapezium: $y = h_1 + (h_2 - h_1) \frac{x}{b}$ $A = \frac{1}{2}b(h_1 + h_2)$ $x_C = \frac{1}{3}b \frac{h_1 + 2h_2}{h_1 + h_2}$